Modelling the Efficiency of a Sheep Pasture in *Minecraft*

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Minecraft is an open-world sandbox video game[1]. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep.

Players can gather large amounts of wool by raising a flock of sheep in a pasture (Figure 1). When a sheep is sheared, they must eat grass to regrow their wool. When a grass block is eaten, it converts into dirt, and can regrow if grass spreads from an adjacent grass block.

One may be tempted to pack the pasture as densely as possible to maximize space efficiency. However, if a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. On the other hand, if the pasture is not densely packed, wool production will still be low because there are not many sheep. Is there some density between these two extremes that will maximize the efficiency of the pasture? We will attempt to answer this question by constructing mathematical models for the sheep and grass populations in the pasture.

1 Deterministic vs. Stochastic

Models of biological populations can be broadly divided into *deterministic* and *stochastic*. Deterministic models are simpler to analyze, but stochastic models are generally more informative since the biological populations, both in the real-world and in *Minecraft*, are stochastic processes. In the real world, populations are influenced by random ecological processes, and in *Minecraft*, the growth of grass and the appetite of sheep are governed by random number generation.

We will begin with a deterministic approach to modelling a pasture, and then move to a stochastic model.

[1] This investigation was conducted in *Minecraft* Java edition version 1.19.4.

Figure 1: A pasture in a meadow, with woolly and sheared sheep, and grass and dirt blocks.

2 Deterministic model

2.1 Grass mechanics

The *Minecraft* physics engine operates in discrete increments of time, called *ticks*. Every tick, a sheep has a chance to attempt to eat the block it's standing on. If it's standing on grass, it will successfully eat the grass and convert it into dirt. If the sheep and grass in the pasture are uniformly distributed, the expected number of sheep standing on grass is the product of the number of sheep and the fraction of grass blocks in the pasture. The rate at which grass is eaten is therefore given by:

$$\mu = \frac{NRN_g}{A} \tag{1}$$

Where μ is the rate at which grass is eaten, A is the size of the pasture, N_g is the number of grass blocks in the pasture, N is the number of sheep, and R is the frequency with which an individual sheep attempts to eat.

The rate at which grass grows not as simple. If N_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If N_g is large, grass growth will be limited by a lack of places for the grass to spread *to*. There are a number of possible models which could be applied here. Let's assume the grass regrowth rate, λ, has a parabolic dependence on N_g:

$$\lambda = kN_g - jN_g^2$$

The growth rate must be zero for $N_g = 0$ and $N_g = A$, which requires $j = \frac{k}{A}$:

$$\lambda = kN_g - \frac{k}{A}N_g^2 \tag{2}$$

This leaves k as a constant related to the rate at which grass spreads. The overall rate of change of N_g is the difference of λ and μ:

$$\begin{aligned} \dot{N}_g = \lambda - \mu &= kN_g - \frac{k}{A}N_g^2 - \frac{NR}{A}N_g \\ &= \left(k - \frac{NR}{A}\right)N_g - \frac{k}{A}N_g^2 \end{aligned} \tag{3}$$

This model, in which $\dot{N}_g$ has quadratic dependence on N_g, is known as the *logistic* model. This model is one of the oldest models for population growth, originally proposed in 1838 [4], and also one of the simplest. It has since been improved upon by more sophisticated models [5], and seldom sees use in real-world ecology, but since we are modelling a video game, it is acceptible.

The analytic solution of the logistic model [5] is:

$$N_g(t) = \frac{P}{1 + me^{-at}} \tag{4}$$

Where

$$a = \left(k - \frac{NR}{A}\right) \qquad P = A - \frac{NR}{k} \qquad m = \frac{P}{N_g(0)} - 1$$

Of note here is that the population asymptotically approaches P as $t \to \infty$. In ecology, P is known as the carrying capacity, the maximum population that a habitat can sustain. We will refer to P as the equilibrium grass population, to avoid confusion with A, which is the number of blocks in the pasture and therefore the maximum number of grass blocks it can contain.

2.2 Experimental determination of rate constants

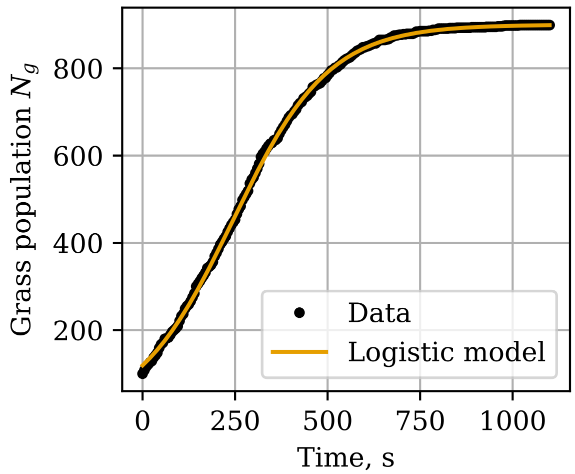
Before proceeding further, it's worth determining the actual values of k and R. This will be done via in-game experiments, which will be described briefly here. More detail can be found on the author's github[2].

Ten sheep were locked in cells with a grass block immediately underneath them. The cells included circuitry to count when each sheep ate the grass and automatically replenish it. In one hour, the sheep consumed 320 blocks of grass. This gives an R value of 0.00888 s^{-1}.

A 30×30 patch of dirt was isolated from its surroundings and uniformly interspersed with 100 grass blocks. As the grass grew, the grass population was recorded every 5 seconds. The data was fit to equation 3, with N fixed at 0. The best fit gave $k = 0.00773\ s^{-1}$. The data and fit are compared in Figure 2. The close fit gives us also confidence that the logistic model is appropriate.

[2] github.com/ELFREELAND/minecraft_sheep

Figure 2: Logistic model fit to grass growth data.



2.3 The fast farmer

For simplicity, let's imagine the farmer has superhuman speed, and is able to shear each sheep the instant it grows its wool. In this case, the rate at which wool is gathered is equal to μ. Denoting the rate of wool gathering with F, and substituting $N_g = P$ into equation 1 (assuming the pasture has reached equilibrium):

$$F_{fast} = \mu = NR - \frac{N^2R^2}{Ak} \tag{5}$$

We can find the optimal sheep population by differentiating equation 5 w.r.t. N, setting the derivative to 0 and solving for N:

$$N = \frac{Ak}{2R} \tag{6}$$

This is the sheep population which will produce the maximum wool output, if the farmer is sufficiently fast. At the optimal value of N, the rate of wool production is:

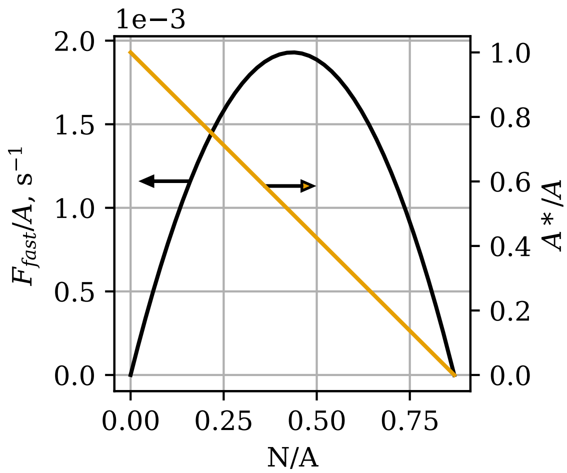
$$F_{fast} = \frac{Ak}{4}$$

Figure 3 shows F_{fast} and P as a function of N, normalized for pasture area. P and F_{fast} are both zero at $\frac{N}{A} = \frac{k}{R} = 0.869$. From a deterministic perspective, the grass in a pasture with $N > 0.869A$ will become extinct.

2.4 The slow farmer

Minecraft players typically don't want to put all of their time into shearing their sheep as quickly as they can. For a more realistic model, we'll assume the

Figure 3: Plot of $\frac{F_{fast}}{A}$ and $\frac{A^*}{A}$ as a function of N, normalized to pasture size.



farmer shears all sheep at once, intermittently and at regular intervals.

Let's define N_w as the number of sheep who have their wool. At time $t = 0$ the farmer shears the sheep, leaving $N_w = 0$, and then goes off to do other things. At time T, they come back to shear the sheep again. The average rate at which the farmer gathers wool is:

$$F_{slow} = \frac{N_w(T)}{T}$$

How many sheep regrew their wool in time T? A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at which sheep are regrowing wool is the rate at which they are eating, μ, times the fraction of sheep that are wool-less. We'll assume the pasture has equilibrated and $N_g = P$:

$$\dot{N}_w = \frac{N - N_w}{N}\mu = (N - N_w)\frac{RP}{A} \tag{7}$$

We can solve equation 7 for $N_w(T)$ by integrating. We'll separate it into N_w terms and non N_w terms:

$$\frac{1}{N - N_w}dN_w = \frac{RP}{A}dt$$

And integrate from initial state ($t = 0, N_w = 0$) to the final state ($t = T, N_w = N_w(T)$):

$$\int_0^{N_w(T)} \frac{1}{N - N_w}dw = \int_0^T \frac{RP}{A}dt$$

$$-\ln\left(\frac{N - N_w(T)}{N}\right) = \frac{TRP}{A}$$

And solve for $N_w(T)$:

$$N_w(T) = N - Ne^{\frac{TRP}{A}}$$

Therefore:

$$F_{slow} = \frac{N - Ne^{-\frac{TRP}{A}}}{T}$$

Finally, we can plug in the expression for P:

$$F_{slow} = \frac{N - Ne^{-TR + \frac{TR^2N}{Ak}}}{T} \tag{8}$$

This is the expression for the rate of wool gathering by a slow farmer. To find the optimal sheep population, we can take the same approach as we did with the fast farmer case. Differentiating (8) w.r.t. N and setting $F_{slow} = 0$:

$$\frac{dF_{slow}}{dN} = \frac{1 - \left(\frac{TR^2N}{Ak} + 1\right)e^{-TR + \frac{TR^2N}{Ak}}}{T} = 0$$

Solving for N yields:

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{9}$$

Where W is the Lambert W function, which satisfies $W(xe^x) = x$. Plugging this back into (8), we obtain the value of F_{slow} at optimal N:

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR + W(e^{TR+1}) - 1})}{R^2T^2} \tag{10}$$

Figure 4 shows a plot of equation 8, and Figure 5 shows plots of equations 9 and 10. As T increases, the optimal sheep population also increases, and the optimal rate of wool gathering decreases. Clearly there is a tradeoff between the frequency with which the farmer shears the flock and the rate at which they gather wool.

There are a couple important things to note about F_{slow}. First, compare the plot of F_{fast} in Figure 3 with the plot of F_{slow} with $T = 1$ in Figure 4 - they are almost identical. We can see this algebraically by applying the following approximation, for $x \ll 1$:

$$e^x \approx x + 1$$

In equation 8, the exponent is small for small values of T, therefore:

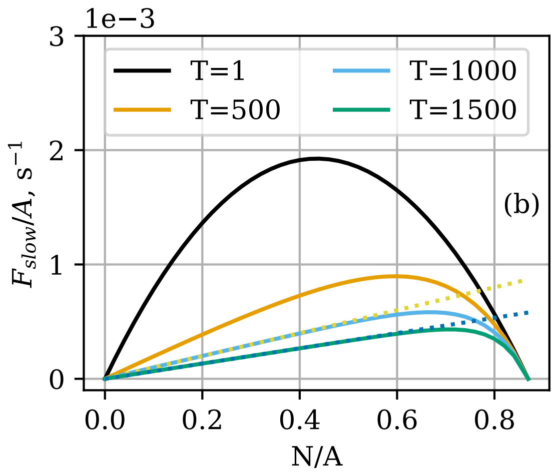
$$F_{slow} \approx \frac{N - N(-TR + \frac{TR^2N}{Ak} + 1)}{T} = NR - \frac{N^2R^2}{Ak} = F_{fast}$$

Second, it appears in Figure 4b that for large values of T and modest values of N, F_{slow} is approximately linear w.r.t. N. Under these conditions, the exponent $-TR + \frac{TR^2N}{Ak}$ in equation 8 is large and negative, and:

$$F_{slow} \approx \frac{N-0}{T} = \frac{N}{T}$$

Physically, this represents conditions for which all of the sheep have regrown their wool by the time the farmer comes back to harvest, so the farmer is simply gathering N blocks of wool with every harvest. Lines of $\frac{N}{T}$ are plotted over top of the curves in Figure 4.

Figure 4: Contours of F_{slow} as a function of N at constant T.



3 Stochastic model

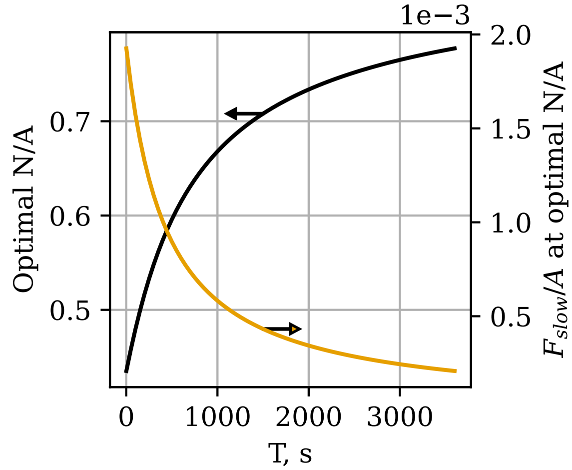
The deterministic model gives valuable insight into the expected behavior of the pasture, but a more complete picture can be obtained by developing a stochastic model which captures the randomness inherent in the physics engine of *Minecraft*. We will develop a stochastic model for the grass population in the pasture using Kolmogorov equations. This approach for population modelling is described in more detail in [1], but the key elements will be outlined here.

3.1 Grass Mechanics

In the deterministic model, we considered the grass population as a single number, N_g, which evolved with time. In the stochastic approach, we will consider the evolution of the *probability distribution* of N_g with time.

We'll introduce the following notation:

Figure 5: Optimal sheep population and maximum wool production as a function of T, normalized to pasture size.



$$p_x(t) = \Pr(N_g(t) = x)$$

In words, $p_x(t)$ is the probability that the pasture has $N_g = x$ at time t.

We'll also need to slightly reconsider the meanings of μ and λ. In the deterministic model, these are the rate of grass growth or consumption. For a small time interval Δt, the number of grass blocks that are eaten is $\mu\Delta t$, and likewise for λ. If Δt is sufficiently short so that the probability of two independent changes in grass population is negligible, $\mu\Delta t$ can be reinterpreted as the probability of one grass block being consumed in time Δt, and likewise for λ. μ and λ therefore represent the probability of the pasture transitioning to a state with one more or one less grass block than before.

Using these new interpretations of μ and λ, we can write a system of differential equations describing the evolution of the probability distribution of N_g with time. Consider a pasture at time $t = t_0$, with $N_g(t_0) = x$. A short time ago, at time $t = t_0 - \Delta t$, the pasture had $N_g(t_0 - \Delta t)$. The pasture may have arrived at $N_g(t_0) = x$ in one of four ways:

1. $N_g(t_0 - \Delta t) = x$, and neither eating nor growth take place.
2. $N_g(t_0 - \Delta t) = x - 1$, and one grass block regrows.
3. $N_g(t_0 - \Delta t) = x + 1$, and one grass block is eaten.
4. Two or more growth and eating events occur.

The probability of each pathway is the probability that the pasture was in the required state $N_g(t_0 - \Delta t)$ times the probability of the event (eating or growth) required to bring the pasture to $N_g(t_0) = x$. The probability of one grass growth in time Δt is $\lambda\Delta t$, and the probability of one grass block being eaten is $\mu\Delta t$. This gives the probability of nothing happening as $(1-\lambda-\mu)\Delta t$. Finally, the probability of multiple events is some product of $\lambda\Delta t$ and $\mu\Delta t$, and is therefore dependent on Δt with order 2 or higher. The probability of each path is therefore:

$$\begin{aligned}
\Pr(1) =& p_x(t_0 - \Delta t)(1-\lambda-\mu)\Delta t \\
\Pr(2) =& p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\
\Pr(3) =& p_{x+1}(t_0 - \Delta t)\mu\Delta t \\
\Pr(4) \approx& O(\Delta t^2)
\end{aligned}$$

The probability of the pasture arriving at $N_g(t_0) = x$ is the sum of the probability of each pathway:

$$\begin{aligned}
p_x(t_0) = & \, p_x(t_0 - \Delta t)(1-\lambda-\mu)\Delta t \\
& + p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\
& + p_{x+1}(t_0 - \Delta t)\mu\Delta t \\
& + O(\Delta t^2)
\end{aligned}$$

subtracting $p_x(t_0 - \Delta t)$ and dividing by Δt:

$$\begin{aligned}
\frac{p_x(t_0) - p_x(t_0 - \Delta t)}{\Delta t} = & -p_x(t_0 - \Delta t)(\lambda + \mu) \\
& + p_{x-1}(t_0 - \Delta t)\lambda \\
& + p_{x+1}(t_0 - \Delta t)\mu \\
& + O(\Delta t^1)
\end{aligned}$$

and taking the limit as $\Delta t \to 0$:

$$\dot{p}_x(t) = -p_x(t)(\lambda(x) + \mu(x)) + p_{x-1}(t)\lambda(x-1) + p_{x+1}(t)\mu(x+1) \tag{11}$$

Note that $\mu(x)$ simply represnts the value of μ with $N_g = x$, and likewise for λ. Note also that the $O(\Delta t^2)$ term has vanished upon taking the limit $\Delta t \to 0$.

Systems of equations such as these, which describe the evolution of a probability distribution with time, are known as *Kolmogorov equations*. We will have a system of $A+1$ equations, one for each possible value of N_g.

We'll wrap up the system into vector form for convenient expression. Let $\mathbf{p}$ be a vector of p_x, and $\dot{\mathbf{p}}$ its derivative:

$$\mathbf{p}(t) = [p_0(t), p_1(t), p_2(t) \dots p_A(t)]$$

$$\dot{\mathbf{p}}(t) = \frac{d\mathbf{p}}{dt} = [\dot{p}_0(t), \dot{p}_1(t), \dot{p}_2(t) \dots \dot{p}_A(t)]$$

Let $\mathbf{R}$ be a matrix of coefficients relating $\mathbf{p}$ and $\dot{\mathbf{p}}$:

$$\mathbf{R} = \begin{bmatrix} r_{0,0} & r_{0,1} & r_{0,2} & \cdots & r_{0,A} \\ r_{1,0} & r_{1,1} & r_{1,2} & \cdots & r_{1,A} \\ r_{2,0} & r_{2,1} & r_{2,2} & \cdots & r_{2,A} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{A,0} & r_{A,1} & r_{A,2} & \cdots & r_{A,A} \end{bmatrix}$$

The coefficient $r_{i,j}$ relates p_i with $\dot{p}_j$:

$$\dot{p}_j = \sum_i r_{ij} p_i$$

$$r_{i,j} = \begin{cases} r_{j-1,j} = \lambda(j-1) \\ r_{j+1,j} = \mu(j) \\ r_{i,i} = -(\lambda(i) + \mu(i)) \\ 0 \ \text{otherwise} \end{cases}$$

Then:

$$\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{R}$$

All that's left to do now is to pick values for N and A and an initial value for N_g and solve. This is most easily done numerically, as has been done here using the Runge-Kutta method. *Minecraft* pastures typically begin fully covered with grass, so we will assume $\mathbf{p}(0) = [0, 0, 0 \dots 1]$.

Figure 6 shows the expected value of N_g after 10000 seconds. We can already see the stochastic model deviating from the deterministic model. At values of N close to $\frac{Ak}{R}$, the expected grass population density is lower than predicted by the deterministic model.

Let's look in a little more detail at the behavior of the pasture. Figure 7 shows the evolution of the distribution of N_g for $N = 40$ and $N = 70$ in a 100-block pasture. Both pastures begin with $N_g = A$, i.e. $\mathbf{p} = [0, 0, 0, \dots 0, 1]$. The pasture with $N = 40$ approaches a stable distribution, evidenced by the indistinguishable distributions at $t = 1600$ s and $t = 9500$ s. In contrast, the pasture with $N = 70$ continues evolving toward smaller and smaller grass populations, with p_0 increasing dramatically as time proceeds.

The reason for this isn't hard to see by considering the equation for $\dot{p}_0$. Beginning with (11) and considering that $p_{-1} = 0$ and $\lambda(0) = \mu(0) = 0$:

$$\dot{p}_0(t) = p_1(t)\mu(1)$$

Since p_1 and $\mu(1)$ are always positive, p_0 is monotonically increasing, bounded above by $p_0 = 1$. Therefore the true equilibrium state is $\mathbf{p} = [1, 0, 0, 0, \dots]$ i.e.

Figure 6: Expected value of N_g at $t = 10000$, for $A = 100$ and $A = 500$. A^* from the deterministic model is shown as a dotted line.

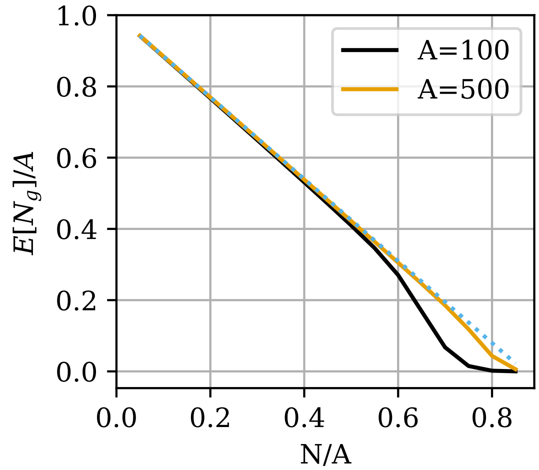
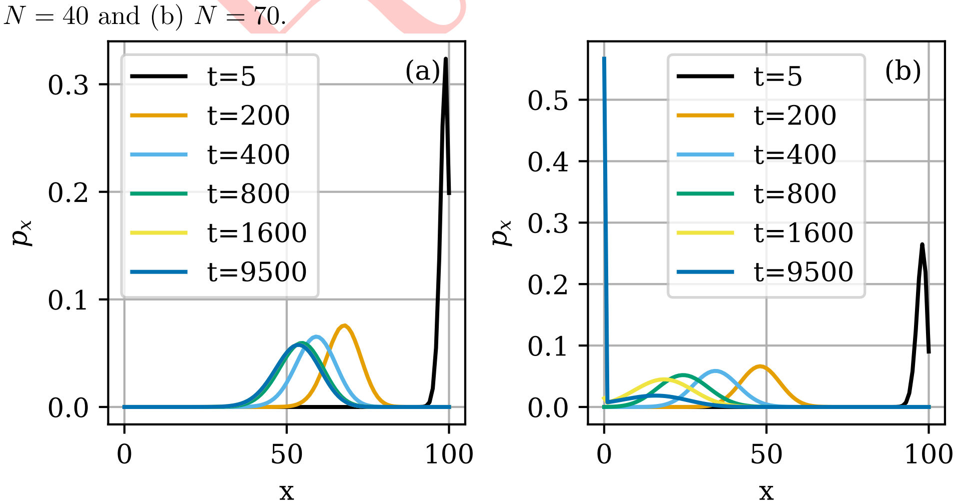


Figure 7: Evolution of N_g distribution for a pasture of size $A = 100$, for (a) $N = 40$ and (b) $N = 70$.



the grass has become exinct. However, even if this is the equilibrium state, the pasture may remain at a different *metastable* (or *quasi-equilibrium*) distribution if the probability of extinction remains negligible across reasonable time scales. This is what is seen in figure 7a. The distribution for $N = 40$ "stabilizes" with a value of p_1 so low that $\dot{p}_0$ remains negligible. The metastable distribution can be also approximated analyically [3] (see also [1] section 6.8.2).

Rapid grass extinction at high sheep densities poses a problem, since the slow farmer deterministic model predicts that the optimal N increases as T increases (see figure 5). Fortunately, an exact solution can be obtained for the expected time to extinction ETE. Let $\mathbf{M}$ be a matrix of mean residence times, such that the element m_{ij} of $\mathbf{M}$ is the expected amount of time the pasture spends with $N_g = j$ before reaching extinction, given that it began with a population i. It can be shown [2] that $\mathbf{M}$ is obtained by removing the first row and column from $\mathbf{R}$ and taking the negative of the inverse of this modified $\mathbf{R}$ matrix. Denoting the modified $\mathbf{R}$ matrix $\mathbf{R}^\dagger$:

$$\mathbf{M} = -(\mathbf{R}^\dagger)^{-1}$$

The sum of the ith row of $\mathbf{M}$ gives the total time spent with nonzero population before becoming extinct, given the pasture started with $N_g = j$. The typical *Minecraft* pasture begins fully populated with grass, i.e. $N_g(0) = A$. Therefore the value we are interested in is:

$$\text{ETE} = \sum_j m_{Aj}$$

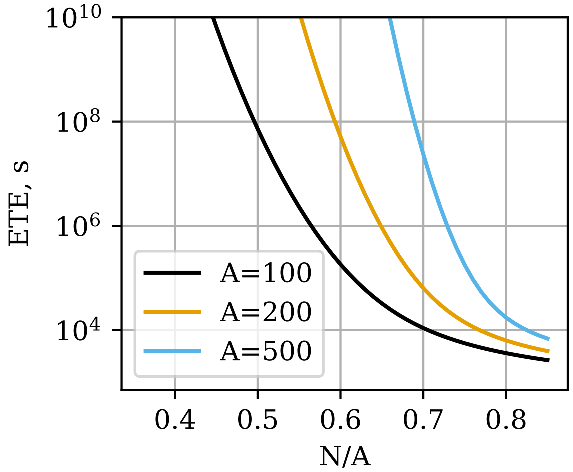
We can apply this solution to obtain the ETE for any value of N and A, as demonstrated in figure 8. For populations close to the deterministic sheep population limit $\frac{N}{A} = \frac{k}{R} = 0.869$, the ETE is on the order of a few hours for all three pasture sizes. As the sheep population density decreases, the ETE increases dramatically - For a 100 block pasture with a sheep density of 0.5, the ETE is on the order of several years. ETE also increases dramatically with larger pastures - A 100-block pasture with sheep density 0.7 has ETE on the order of a few hours, while a 500-block pasture with the same density has ETE on the order of months.

We have seen that pastures with long ETE, such as the one depicted in Figure 7a, approach a metastable distribution as they evolve. This distribution can actually be calculated analytically. The pasture will cease evolving when all $\dot{p}_x = 0$. This implies that the likelihood of the pasture having $N_g = x$ and losing one grass block equals the probability of $N_g = x - 1$ and growing one grass block [3]. In symbols:

$$p_x \mu(x) = p_{x-1} \lambda(x-1) \tag{12}$$

We will assume $N_g = 0$ is an unreachable state and exclude it from consideration. Writing (12) for $2 \leq x \leq A$ and combining with the normalization condition, we obtain a system of A equations in A variables:

Figure 8: Expected time to extinction for pastures of $A = 100$, 200, and 500 as a function of sheep population density.



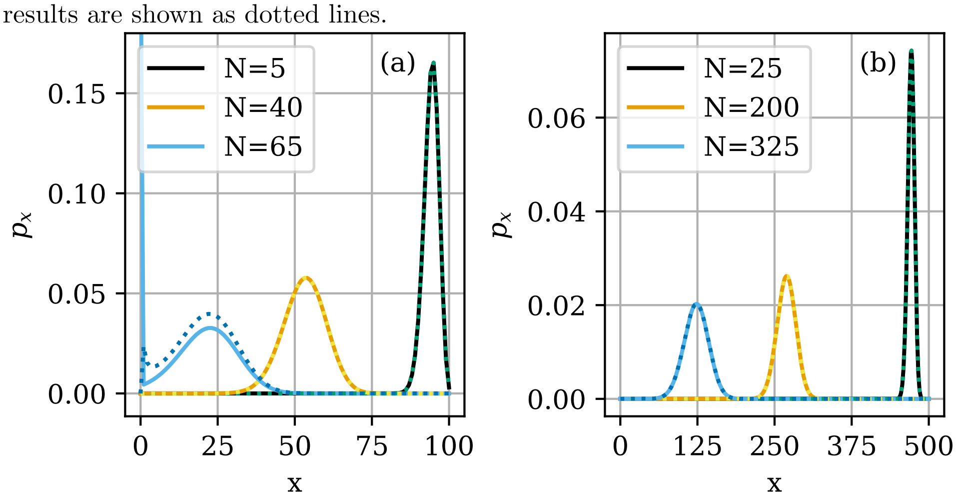
$$\begin{cases} p_2\mu(2) - p_1\lambda(1) = 0 \\ p_3\mu(3) - p_2\lambda(2) = 0 \\ \dots \\ p_A\mu(A) = p_{A-1}\lambda(A-1) = 0 \\ \sum_{x=1}^{A} p_x = 1 \end{cases} \tag{13}$$

We can solve (13) to obtain the metastable N_g distribution. Figure 9 shows the metastable distribution for a few select conditions. The metastable distribution predicted by (13) closely matches the numerical solution of the Kolmogorov equations for $t = 10000$ in all cases except $A = 100$, $N = 65$. In that case, the ETE is not sufficiently long to allow a metastable distribution to exist. (13) also produces unrealistic results in this case - the distribution shows an increase in p_x for small x; such a distribution would not actually be metastable.

3.2 The fast farmer

The mechanics of the fast farmer are rather simple. The expected value of $E[F_{fast}] = E[\mu]$, and since μ is directly proportional to N_g, $E[F_{fast}]$ can be calculated by simply plugging $E[N_g]$ into equation 1. Figure 6 shows that $E[N_g]$ predicted by the Kolmogorov equations does not differ substantially from $A*$ from the deterministic model unless the ETE is low.

Figure 9: Metastable population distributions for pastures of size (a) $A = 100$ and (b) $A = 500$, for three different sheep population densities. Numerical results are shown as dotted lines.



3.3 The slow farmer

It seems natural at this point to pursue an analysis of the slow farmer with Kolmogorov equations. However, this is a daunting task. We have two variables, N_g and N_w, evolving simultaneously. We can construct a system of Kolmogorov equations for this situation, considering each combination of N_g and N_w as a possible state of the pasture. This will lead to a system of $(A + 1)(N + 1)$ equations.

If we want to apply this analysis to the same conditions as the fast farmer, we will have at most $(500 + 1)(425 + 1) = 213426$ simultaneous equations, and a coefficient matrix with as many rows and columns. Such a matrix would occupy 182 GB of memory if stored as 64-bit floats, not to mention, the system of equations is now so large that computation becomes intractable. Some degree of simplification will need to be made, and analysis of the resulting model will be long, so this task is deferred for later.

4 What next?

The logistic model, and the Kolmogorov equations, have proven useful for predicting the behavior of a pasture in *Minecraft*. However, there are a few assumptions we made that leave unanswered questions.

First, time in *Minecraft* is discretized into increments of 0.05 seconds, while we have been modelling the pasture with continuous-time processes. What, if any, effect would this have on the behavior of the pasture, as compared to the behavior predicted by continuous-time models?

Second, a pasture in *Minecraft* is typically constructed by fencing off an area of grass from a larger meadow. Grass can spread under fences, effectively introducing an immigration term to the logistic model that is nonzero even when the grass in the pasture is extinct. Such a model is studied in [1] chapter 7, but the deterministic model becomes rather unwieldy, so the effect of immigration was ignored here. However, introducing immigration would have an important effect on the behavior of the pasture for small N_g, when λ is otherwise small.

References

[1] Matis, James, and Thomas Kiffe, eds. *Stochastic Population Models: A Compartmental Perspective*. 1st ed. Lecture Notes in Statistics. New York: Springer, n.d.

[2] Matis, J. H., T. E. Wehrly, and C. M. Metzler. "On Some Stochastic Formulations and Related Statistical Moments of Pharmacokinetic Models." *Journal of Pharmacokinetics and Biopharmaceutics* 11, no. 1 (1983): 77-92. https://doi.org/10.1007/BF01061769.

[3] Bartlett, M. S., J. C. Gower, and P. H. Leslie. "A Comparison of Theoretical and Empirical Results for Some Stochastic Population Models." *Biometrika* 47, no. 1-2 (June 1, 1960): 1-11. https://doi.org/10.1093/biomet/47.1-2.1.

[4] Verhulst, P. "Notice sur la loi que la population suit dans son accroissement." *Correspondance Mathématique et Physique* 10 (1838): 113-21.

[5] Banks, Robert B. *Growth and Diffusion Phenomena*. Texts in Applied Mathematics. Springer Berlin, Heidelberg, 1994.